



Georgia Tech
College of
Engineering

COE 3001

MECHANICS OF DEFORMABLE BODIES

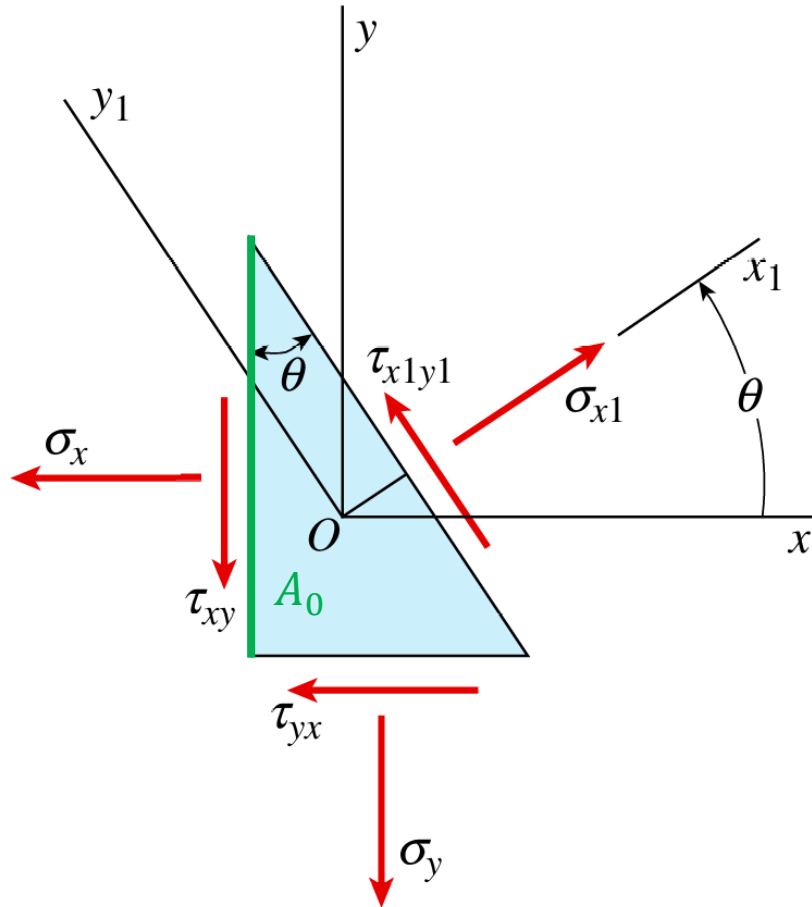
Lecture 19 – Analysis of stress – strain transformation

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Apr. 8, 2026

Stresses on inclined sections



Equilibrium equation in x direction

$$-\sigma_x A_0 - \tau_{xy} A_0 \tan \theta - \frac{\tau_{x1y1} A_0}{\cos \theta} \sin \theta + \frac{\sigma_{x1} A_0}{\cos \theta} \cos \theta = 0$$

Equilibrium equation in y direction

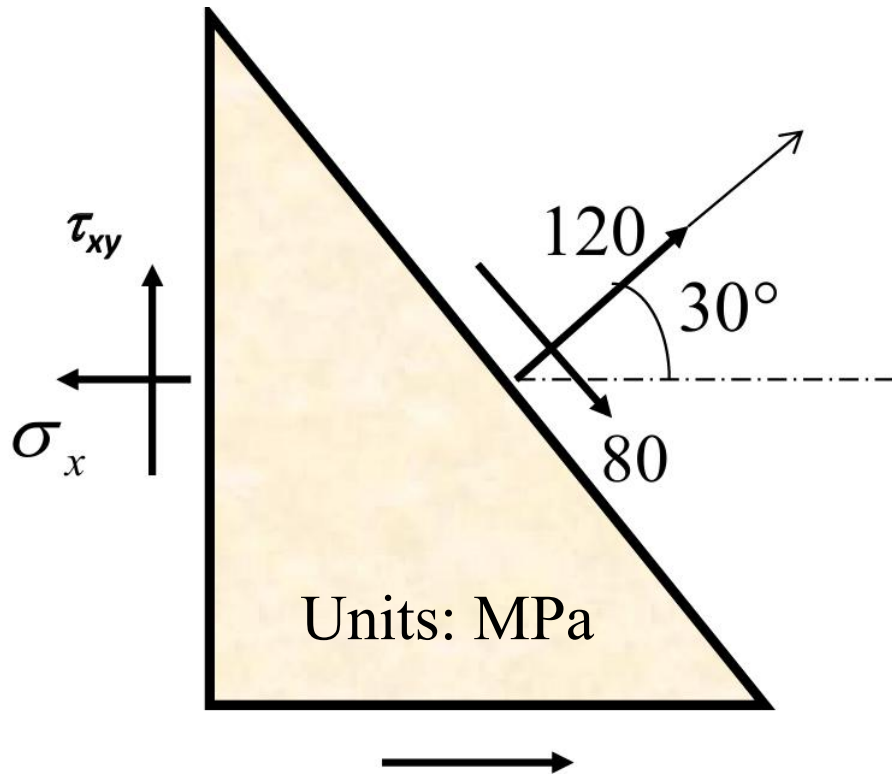
$$-\tau_{xy} A_0 - \sigma_y A_0 \tan \theta + \frac{\tau_{x1y1} A_0}{\cos \theta} \cos \theta + \frac{\sigma_{x1} A_0}{\cos \theta} \sin \theta = 0$$

Solved

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Example



$$\sigma_\alpha = 120\text{MPa}, \quad \tau_\alpha = 80\text{MPa}, \\ \alpha = 30^\circ, \quad \sigma_y = 0$$

$$\begin{aligned} \sigma_\alpha &= \frac{\sigma_x}{2} + \frac{\sigma_x}{2} \cos 60^\circ - \tau_{xy} \sin 60^\circ \\ &= \frac{3}{4} \sigma_x - \frac{\sqrt{3}}{2} \tau_{xy} = 120\text{MPa} \end{aligned}$$

$$\begin{aligned} \tau_\alpha &= \frac{\sigma_x}{2} \sin 60^\circ + \tau_{xy} \cos 60^\circ \\ &= \frac{\sqrt{3}}{4} \sigma_x + \frac{\tau_{xy}}{2} = 80\text{MPa} \end{aligned}$$

$$\sigma_x = 80 + 160 \frac{\sqrt{3}}{3} = 172.4\text{MPa} \quad \tau_{xy} = 80 - 40\sqrt{3} = 10.7\text{MPa}$$

Mohr's circle

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\sigma_{x1} - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\left(\sigma_{x1} - \frac{\sigma_x + \sigma_y}{2} \right)^2 + (\tau_{x1y1})^2 = \left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2$$

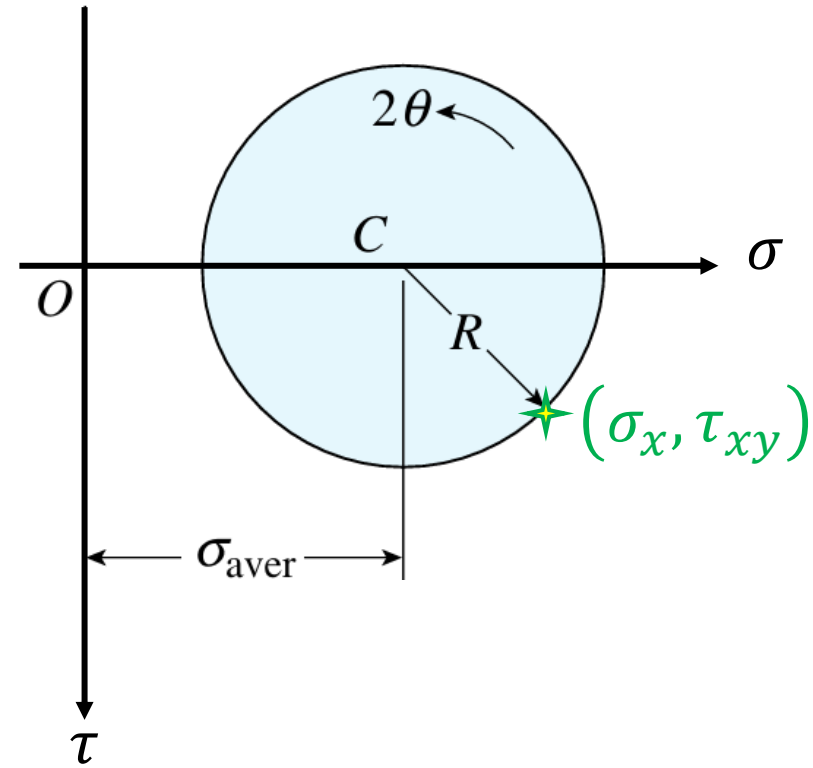
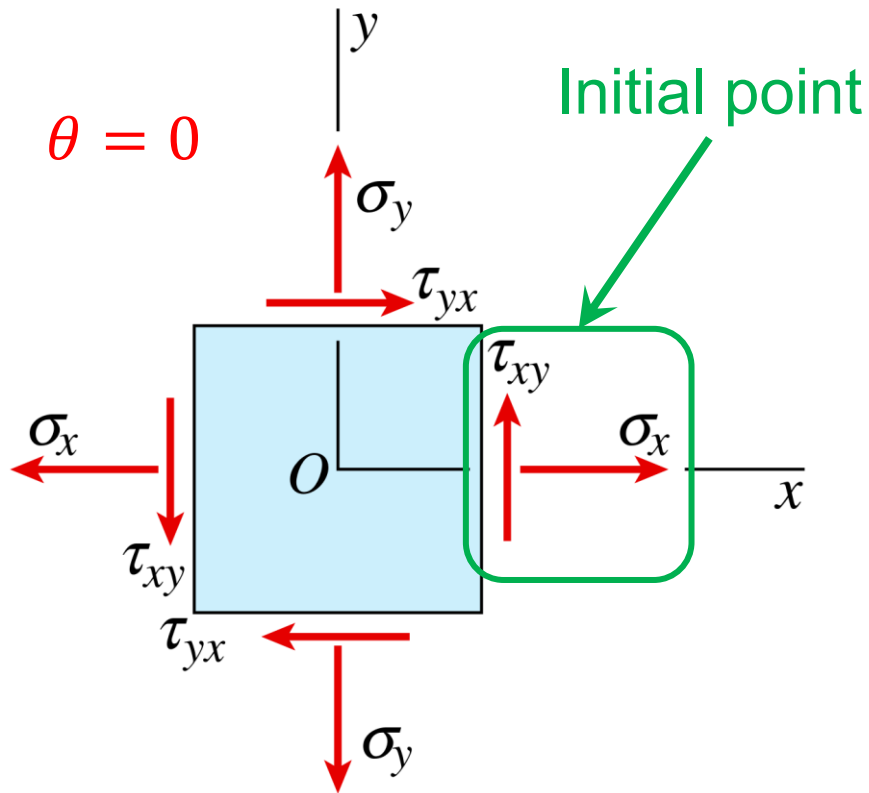
Center:

$$(\sigma_{\text{aver}}, 0) = \left(\frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

Radius:

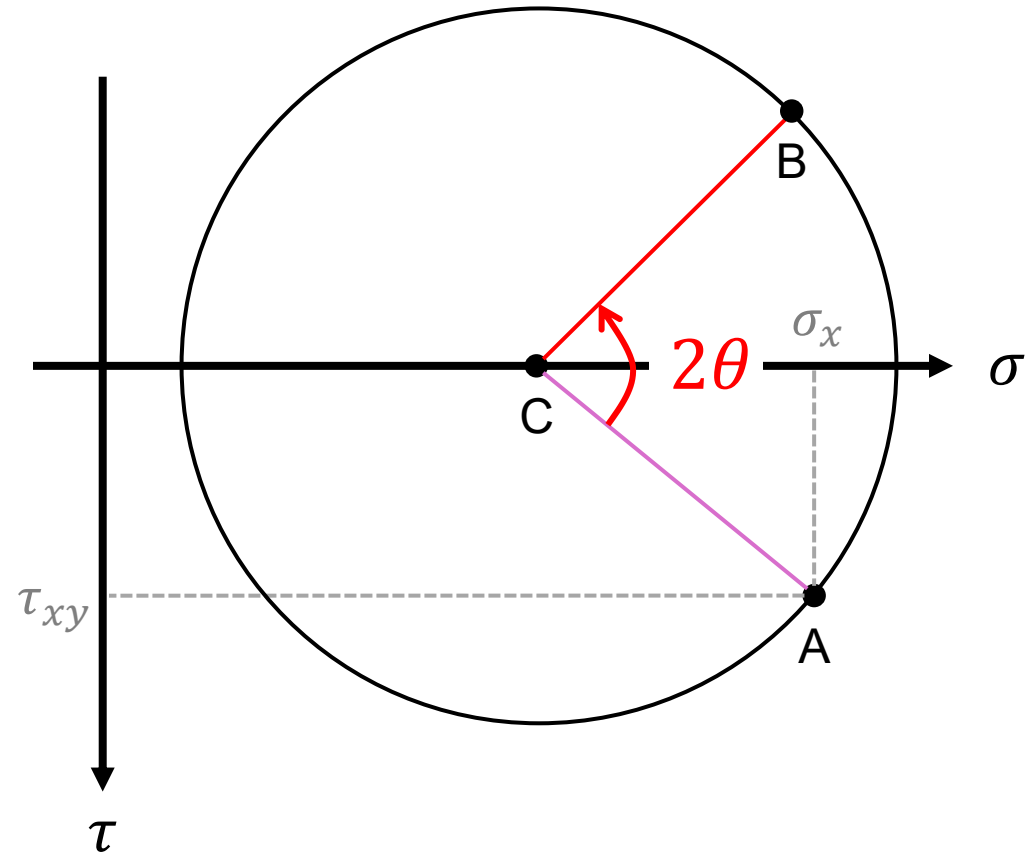
$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

Mohr's circle: initial point



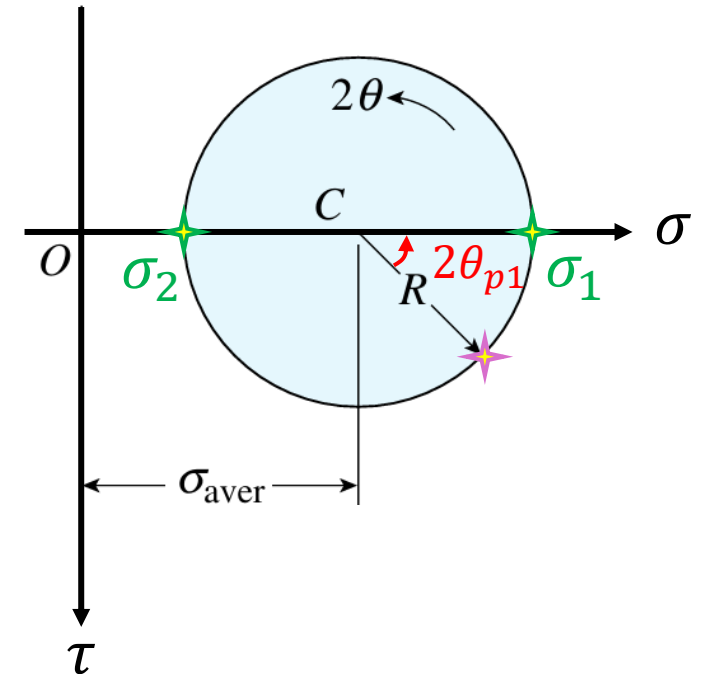
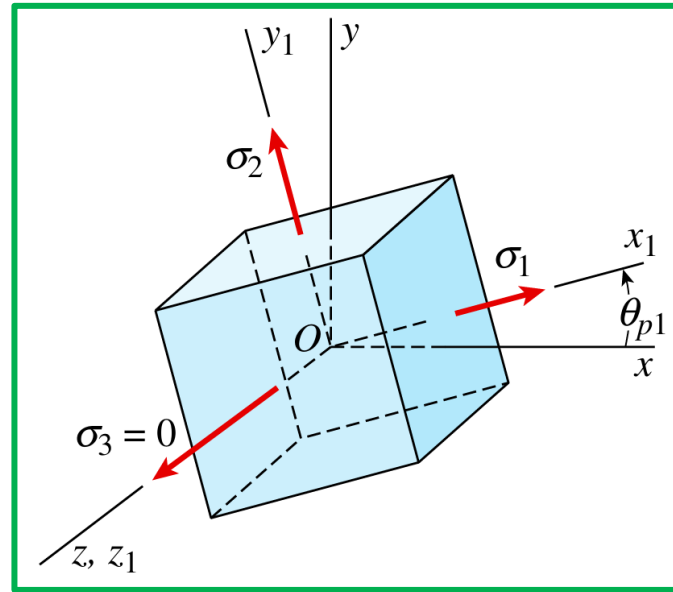
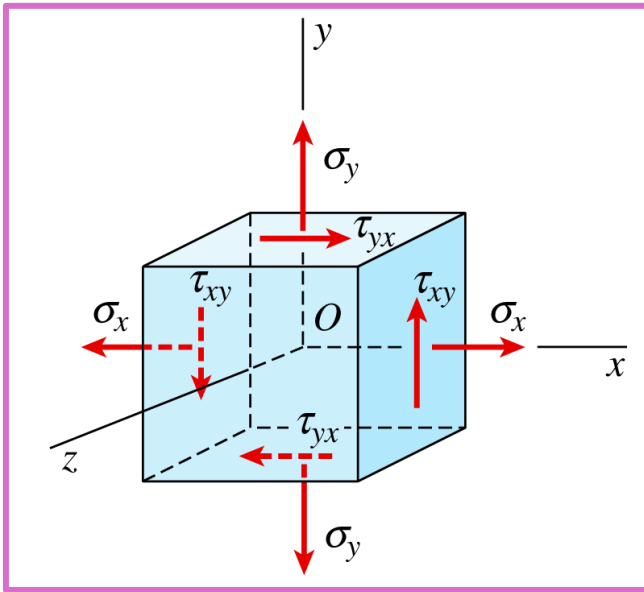
Draw Mohr's circle

- Draw coordinate axes. τ axis positive downward.
- Locate the center C with coordinate $(\sigma_{\text{aver}}, 0)$
- Locate the current stress state point A with coordinate (σ_x, τ_{xy})
- Draw the Mohr's circle with C as center and CA as radius
- Rotate CA by 2θ to get the stress state on inclined plane B

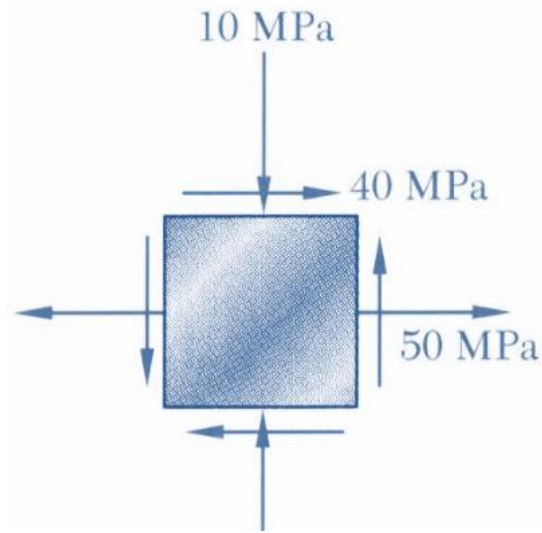


Principal Stresses

Principal stresses ($\sigma_1, \sigma_2, \sigma_1 > \sigma_2$) are the maximum and minimum normal stresses at an angle, where shear stresses are zero.



Example



$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$
$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

To find the principal stress

$$\frac{d\sigma_{x1}}{d\theta} = -(\sigma_x - \sigma_y) \sin 2\theta + 2\tau_{xy} \cos 2\theta = 0$$

We have

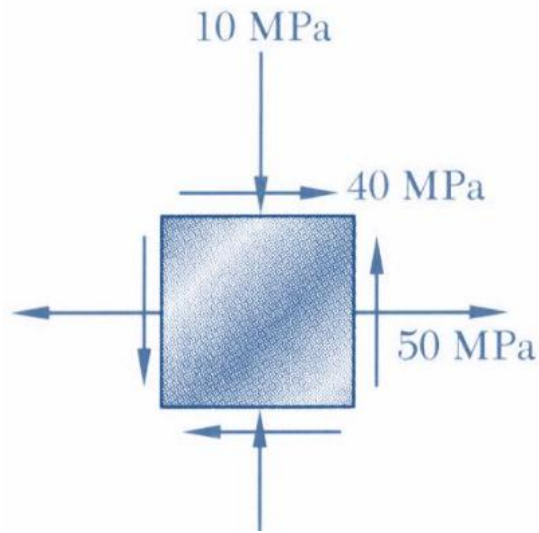
$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$2\theta_p = 53.1^\circ, 233.1^\circ$$

$$\theta_p = 26.6^\circ, 116.6^\circ$$

- For the state of plane stress shown,
- (a) construct Mohr's circle,
 - (b) determine the principal planes
 - (c) calculate the principal stresses
 - (d) calculate the maximum shear stress and the corresponding normal stress

Example



$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

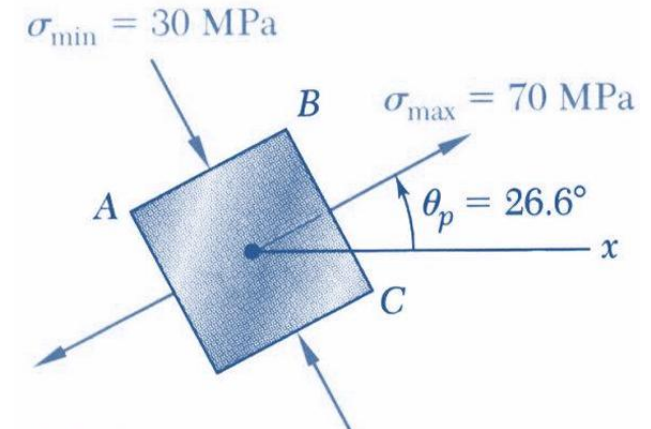
2) Determine the principal stresses from

$$\sigma_{\max, \min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 20 \pm \sqrt{(30)^2 + (40)^2}$$

$$\sigma_{\max} = 70 \text{ MPa}$$

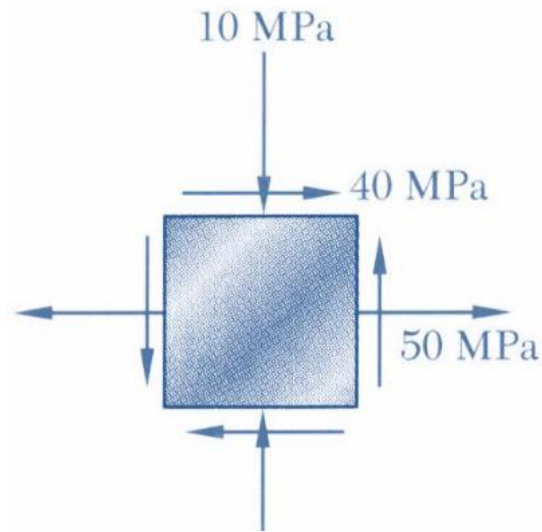
$$\sigma_{\min} = -30 \text{ MPa}$$



For the state of plane stress shown,

- construct Mohr's circle,
- determine the principal planes
- calculate the principal stresses
- calculate the maximum shear stress and the corresponding normal stress

Example



For the state of plane stress shown,
 (a) construct Mohr's circle,
 (b) determine the principal planes
 (c) calculate the principal stresses
 (d) calculate the maximum shear stress and the corresponding normal stress

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x1y1} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

3) Calculate the maximum shearing stress with

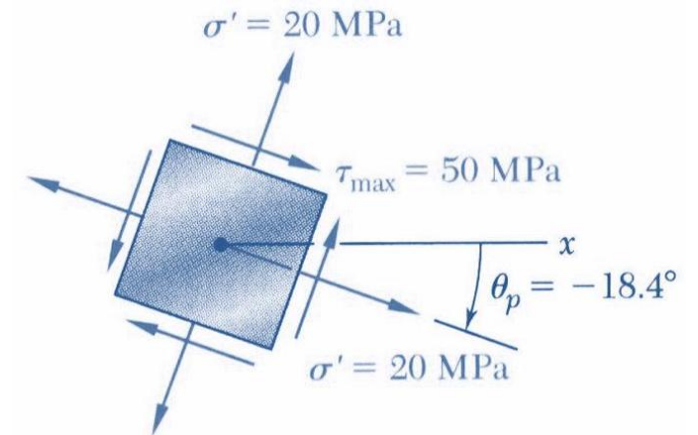
$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$\tau_{\max} = 50 \text{ MPa}$$

$$\theta_s = \theta_p - 45$$

$$\theta_s = -18.4^\circ, 71.6^\circ$$

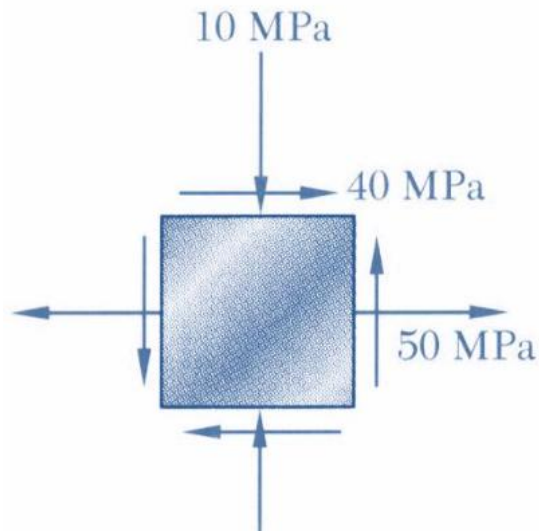


4) The corresponding normal stress is

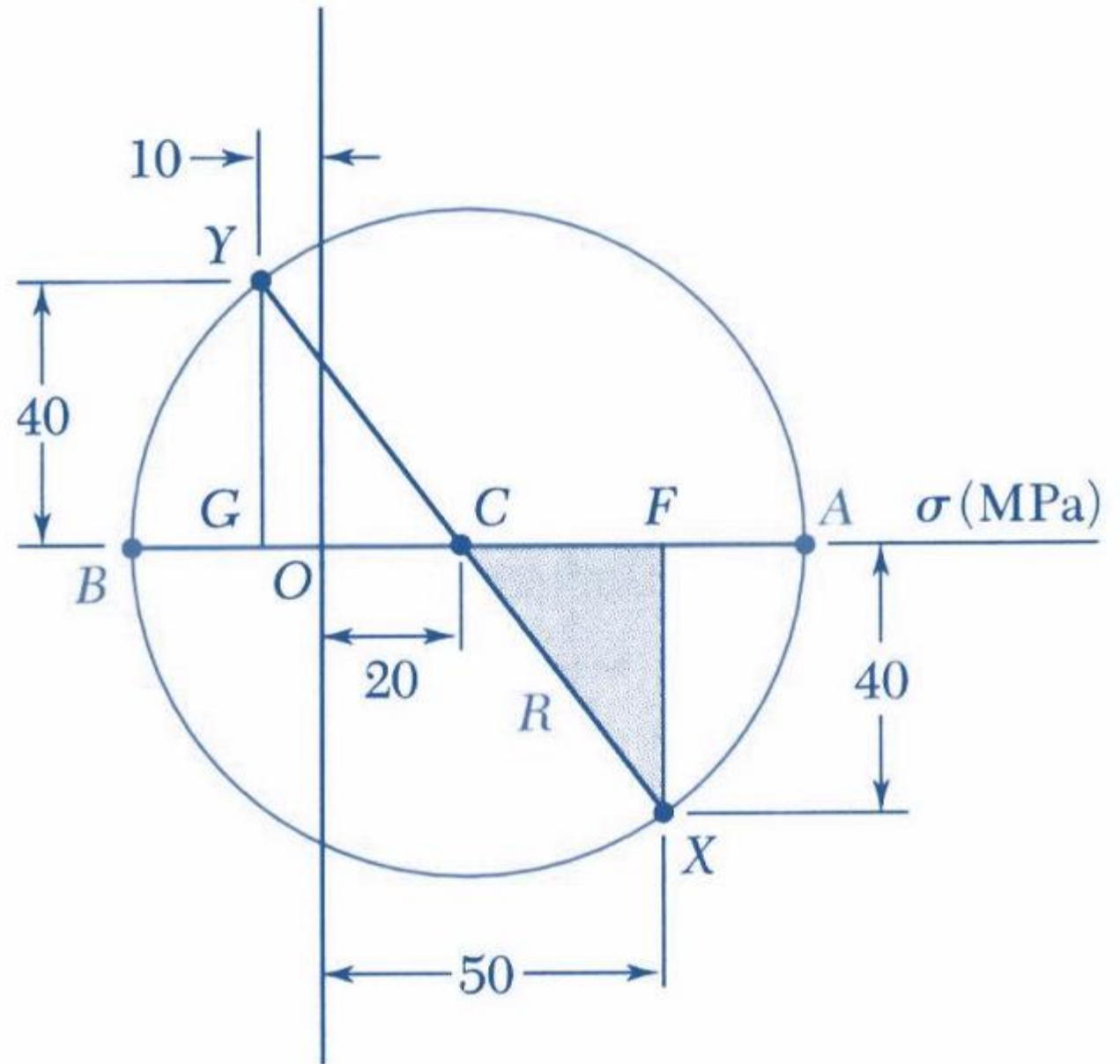
$$\sigma' = \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{50 - 10}{2}$$

$$\sigma' = 20 \text{ MPa}$$

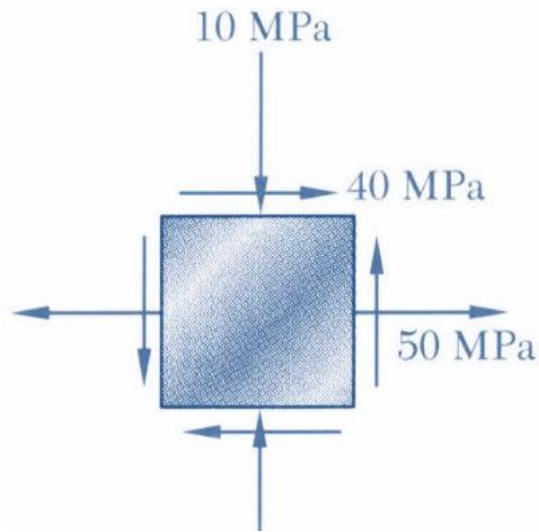
Example



- For the state of plane stress shown,
- (a) construct Mohr's circle,
 - (b) determine the principal planes
 - (c) calculate the principal stresses
 - (d) calculate the maximum shear stress and the corresponding normal stress

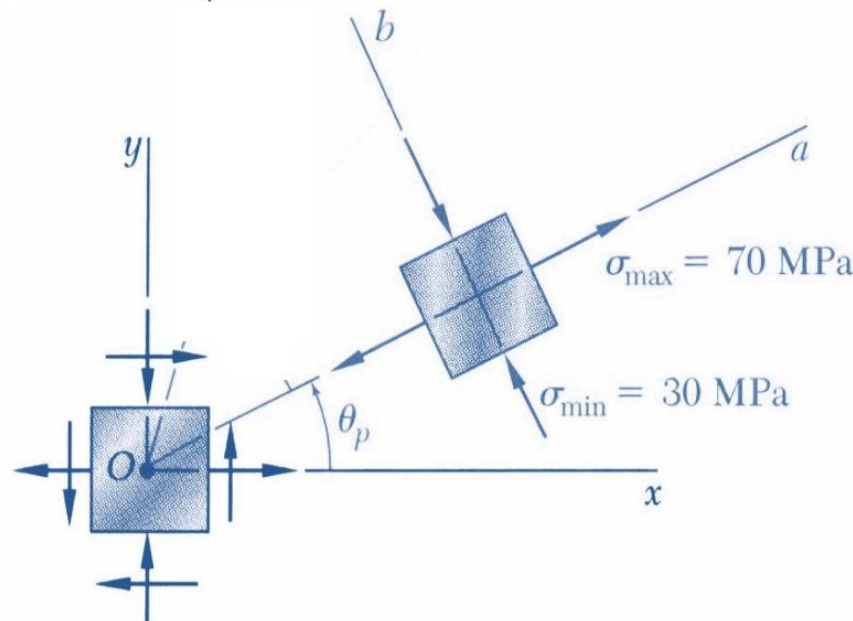
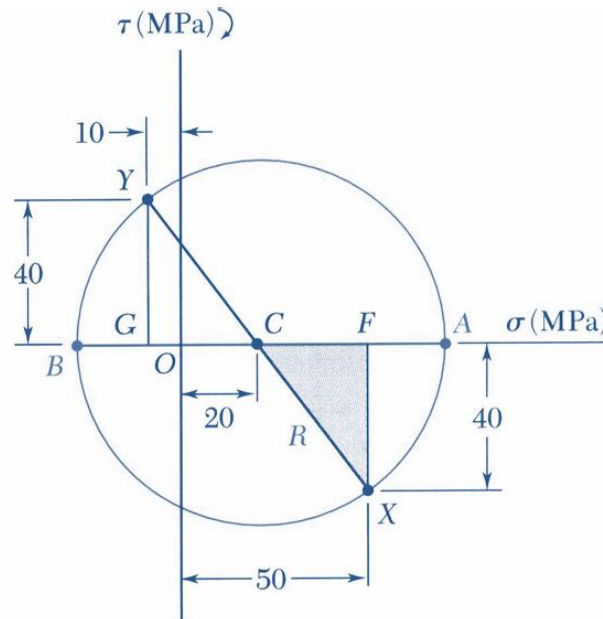


Example



For the state of plane stress shown,

- (a) construct Mohr's circle,
- (b) determine the principal planes
- (c) calculate the principal stresses
- (d) calculate the maximum shear stress and the corresponding normal stress



2) Principal planes and stresses

$$\sigma_{\max} = OA = OC + CA = 20 + 50$$

$$\sigma_{\max} = 70 \text{ MPa}$$

$$\sigma_{\min} = OB = OC - BC = 20 - 50$$

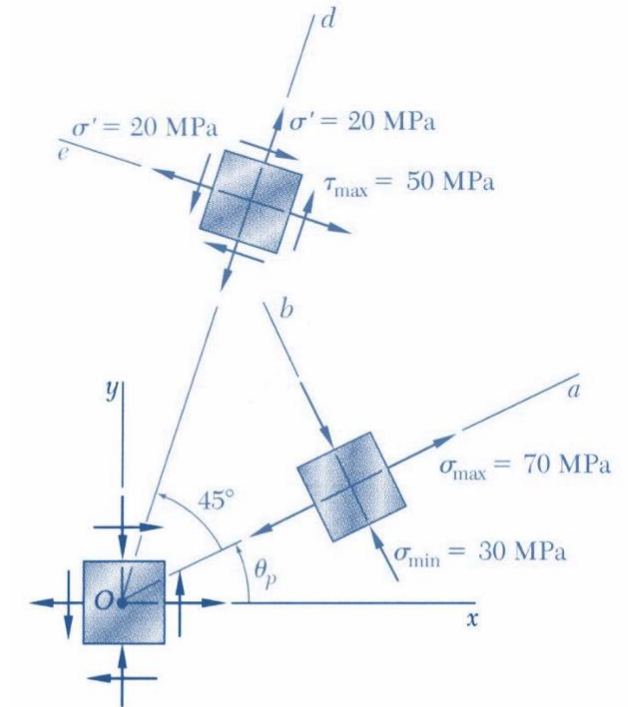
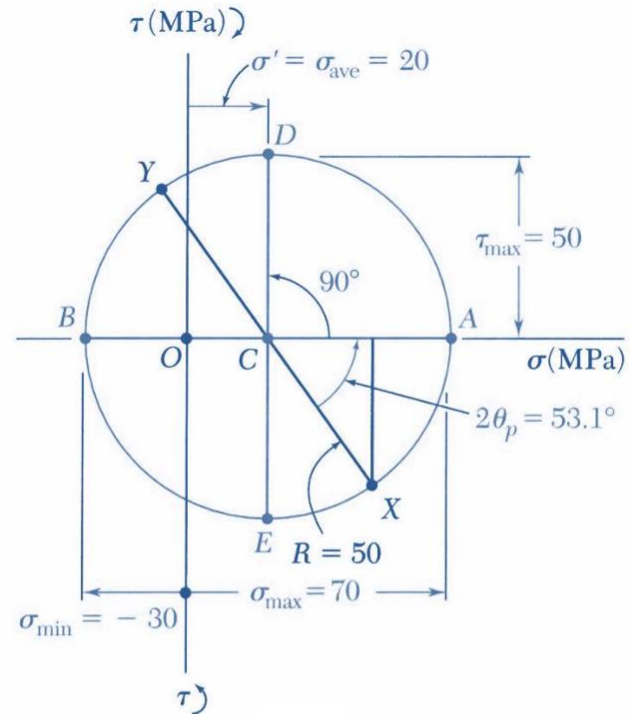
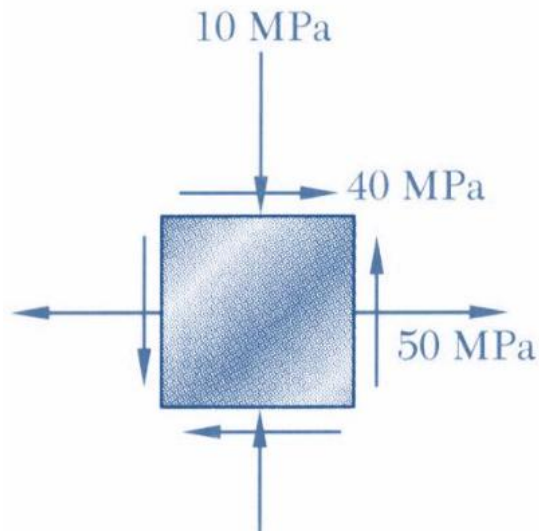
$$\sigma_{\min} = -30 \text{ MPa}$$

$$\tan 2\theta_p = \frac{FX}{CP} = \frac{40}{30}$$

$$2\theta_p = 53.1^\circ$$

$$\theta_p = 26.6^\circ$$

Example



For the state of plane stress shown,

- construct Mohr's circle,
- determine the principal planes
- calculate the principal stresses
- calculate the maximum shear stress and the corresponding normal stress

3) Maximum shear stress

$$\theta_s = \theta_p + 45^\circ$$

$$\theta_s = 71.6^\circ$$

$$\tau_{\max} = R$$

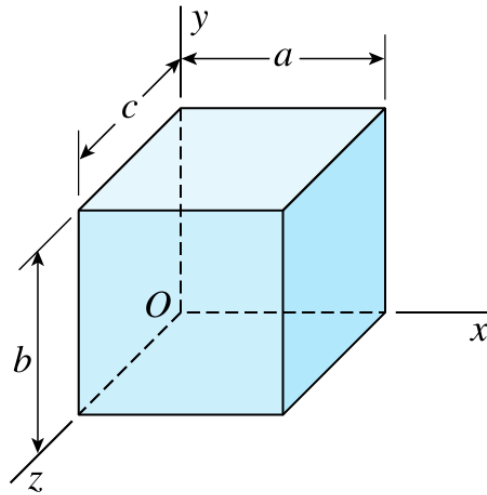
$$\tau_{\max} = 50 \text{ MPa}$$

$$\sigma' = \sigma_{\text{ave}}$$

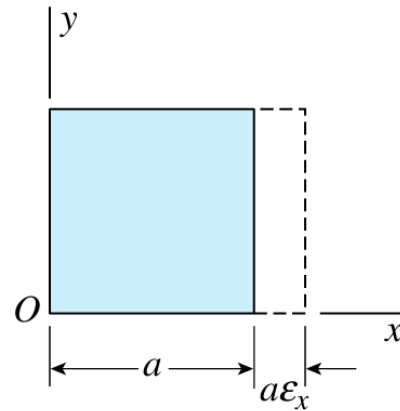
$$\sigma' = 20 \text{ MPa}$$

Plane strain

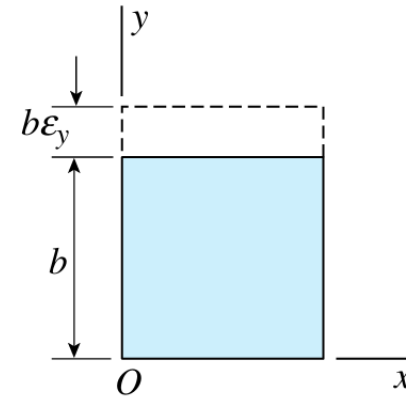
If the only deformations are those in the x - y plane, then three strain components may exist (ϵ_x , ϵ_y , γ_{xy}). An element of material subjected to these strains (and only these strains) is said to be in a state of **plane strain**.



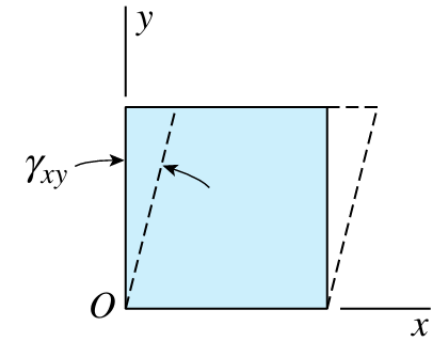
(a)



(b)



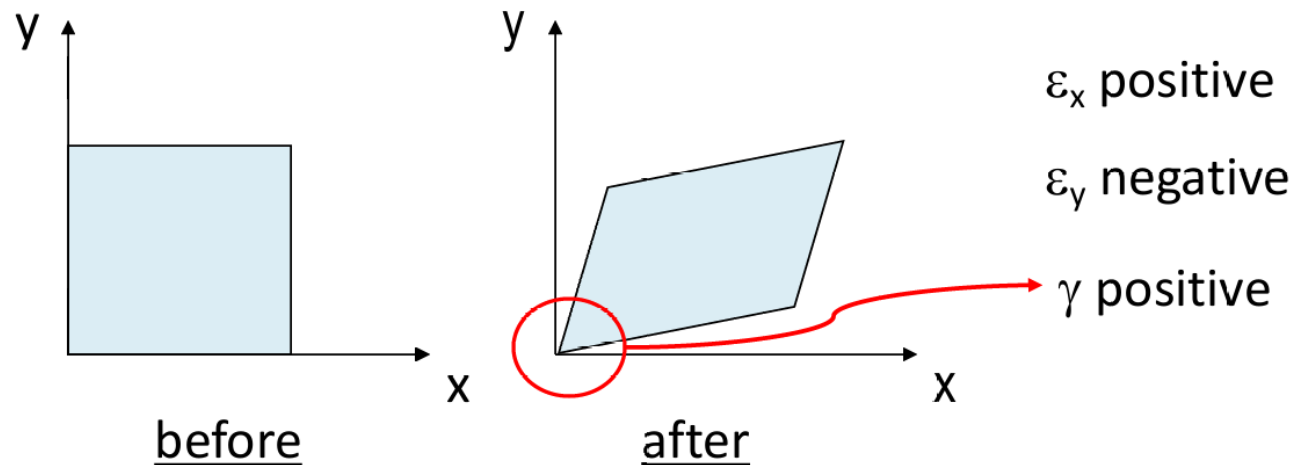
(c)

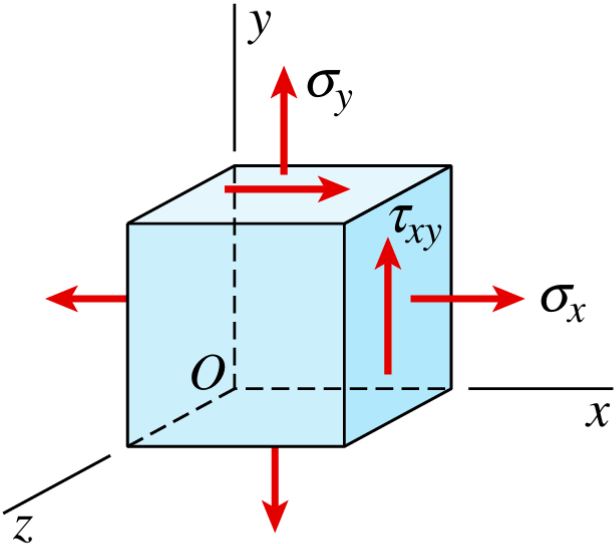
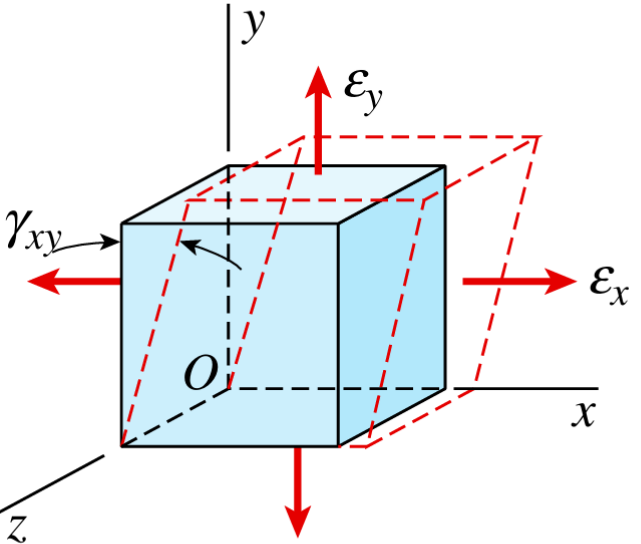


(d)

Sign convention

- A **normal strain ϵ** has a subscript that identifies the direction on which the deformation occurs. **Tension is positive and compression is negative**
- A **shear strain γ** has two subscripts, indicate the plane where the deformation occurs. A shear stress is **positive when the angle between two originally perpendicular line decreases**.



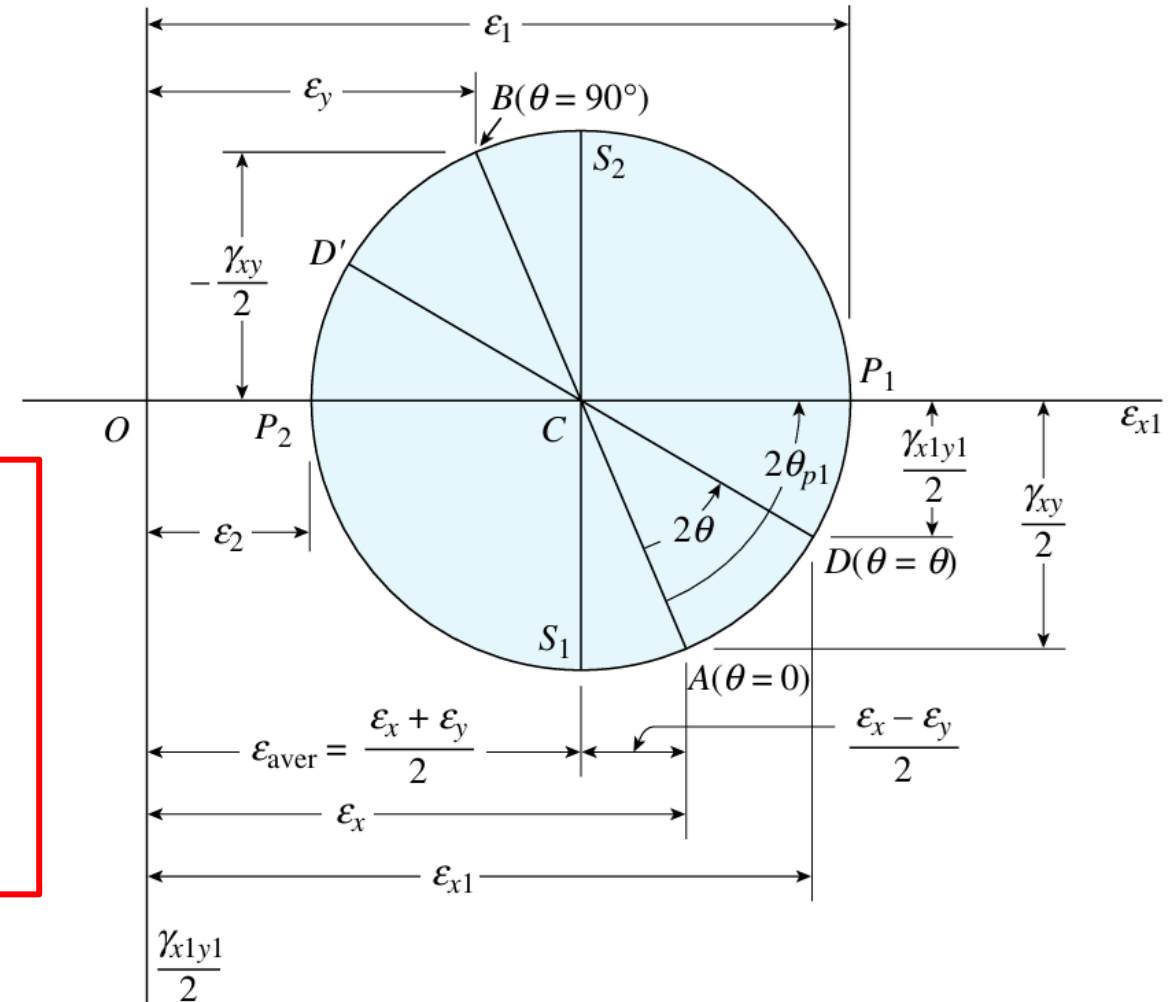
	Plane stress	Plane strain
		
Stresses	$\sigma_z = 0 \quad \tau_{xz} = 0 \quad \tau_{yz} = 0$ $\sigma_x, \sigma_y,$ and τ_{xy} may have nonzero values	$\tau_{xz} = 0 \quad \tau_{yz} = 0$ $\sigma_x, \sigma_y, \sigma_z,$ and τ_{xy} may have nonzero values
Strains	$\gamma_{xz} = 0 \quad \gamma_{yz} = 0$ $\epsilon_x, \epsilon_y, \epsilon_z,$ and γ_{xy} may have nonzero values	$\epsilon_z = 0 \quad \gamma_{xz} = 0 \quad \gamma_{yz} = 0$ $\epsilon_x, \epsilon_y,$ and γ_{xy} may have nonzero values

Transformation equations for plane strain

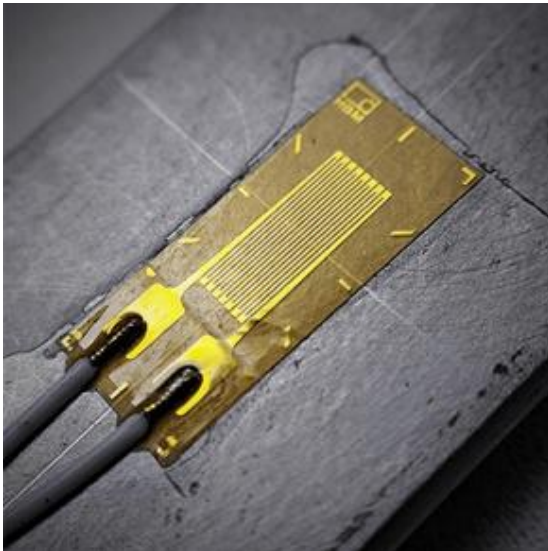
The same format as the stress transformation equations, replacing σ by ε , and τ by $\gamma/2$

$$\varepsilon_{x1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta$$

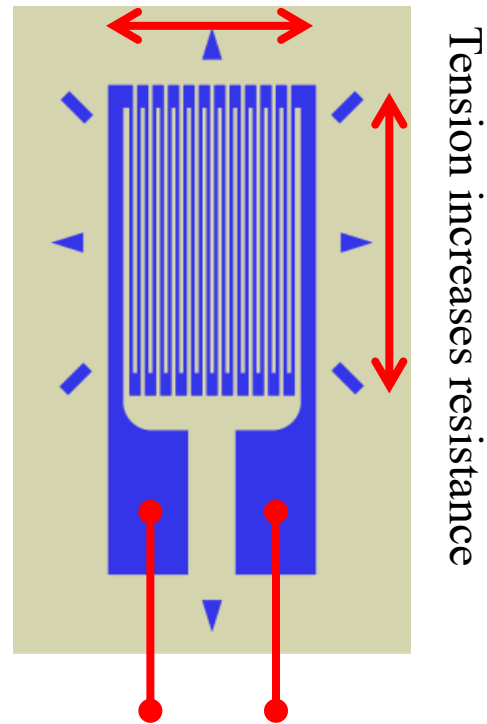
$$\frac{\gamma_{x1y1}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2} \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$$



Strain gauge

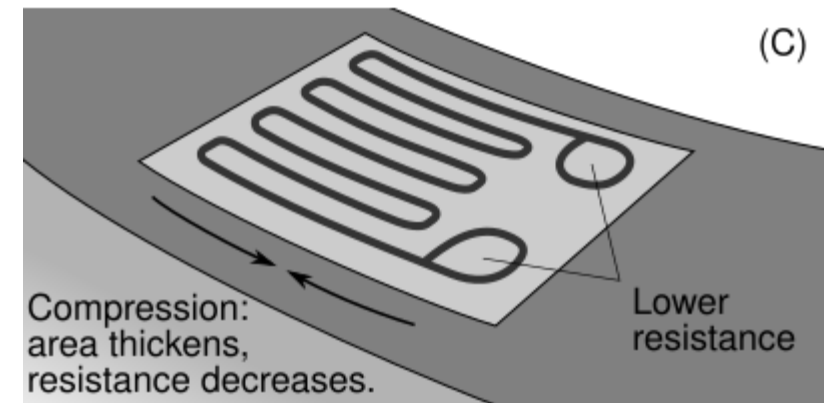
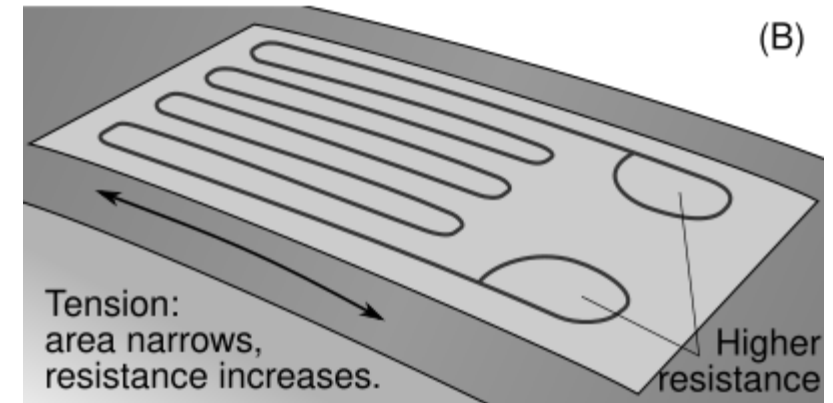
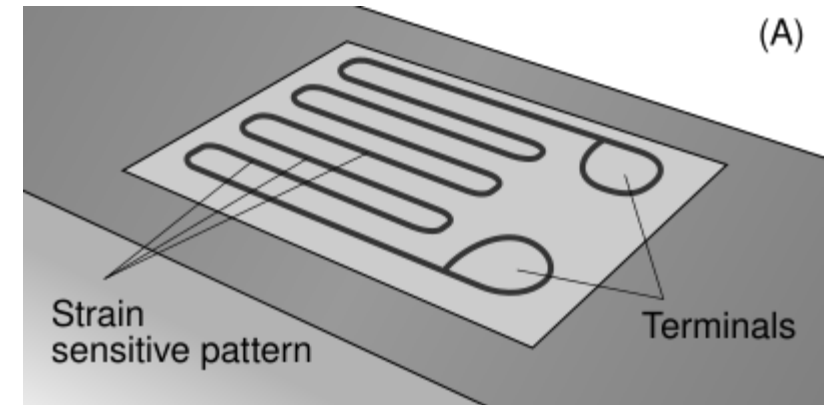


Insensitive to lateral deformation



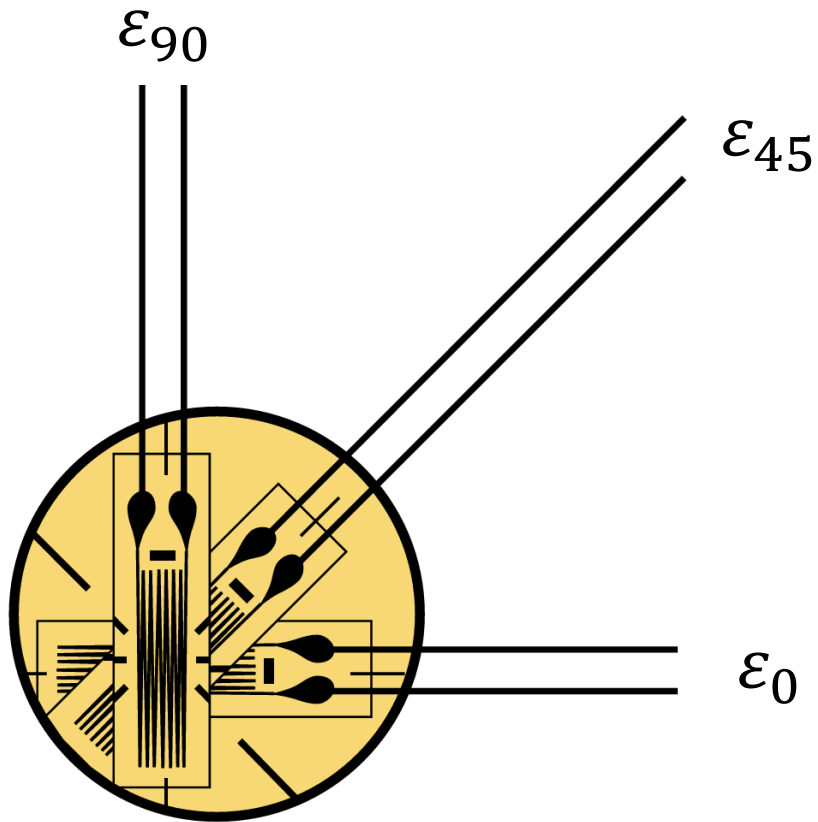
Tension increases resistance

Measured resistance
 $R \propto L$



Example: 45° strain rosette

$$\varepsilon_{x1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta$$
$$\frac{\gamma_{x1y1}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2} \sin 2\theta + \frac{\gamma_{xy}}{2} \cos 2\theta$$



Determine ε_x , ε_y , γ_{xy}

0°:

$$\varepsilon_0 = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} + 0 \Rightarrow \varepsilon_x = \varepsilon_0$$

90°:

$$\varepsilon_{90} = \frac{\varepsilon_x + \varepsilon_y}{2} - \frac{\varepsilon_x - \varepsilon_y}{2} + 0 \Rightarrow \varepsilon_y = \varepsilon_{90}$$

45°:

$$\varepsilon_{45} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \times 0 + \frac{\gamma_{xy}}{2} \times 1$$
$$\Rightarrow \gamma_{xy} = 2\varepsilon_{45} - \varepsilon_x - \varepsilon_y = 2\varepsilon_{45} - \varepsilon_0 - \varepsilon_{90}$$